ITSIMPLERA SOLUTIONS

INTERNSHIP WEEKLY REPORT

FOUNDER: ENGR ZIA ULLAH

WEEK 02: BASELINE MACHINE LEARNING MODELING

Target System: Steel Industry Energy Consumption Optimization

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- Role: AI/ML Intern
- Week: 02
- Instructor / Team Lead: Mr Rayyan Khan
- Submission Date: 11-07-2026

Project Introduction:

Introduction to Baseline Modelling:

What are we doing? We are building a basic computer system to guess how much electricity a steel factory uses based on its daily data.

How are we doing it? We are trying out two simple methods (Linear Regression and a Decision Tree) to see which one makes better guesses.

Why does it matter? We need to start with these simple methods today so that next week, if we build a more advanced system, we can easily prove if it is actually better or not.

Task Overview & Our Dataset:

Our Goal: Use the dataset features to accurately predict the target column: Usage_kWh.

Important columns we look at:

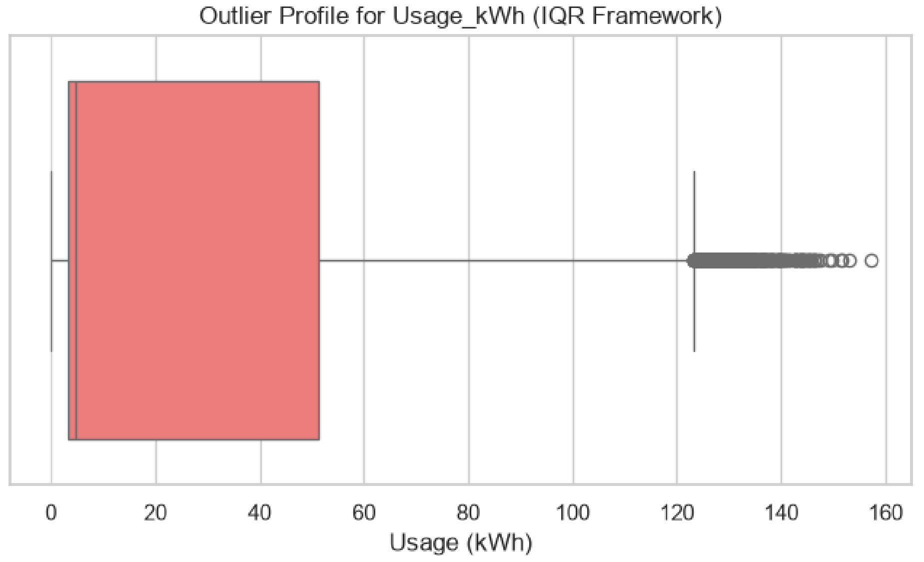
Lagging_Current_Power_Factor and Leading_Current_Power_Factor

CO2(tCO2) (Carbon emissions).

Data Preparation: Cleaned up missing values and split the data so we can train on one part and test on another

[2]:

	date	Usage_kWh	Lagging_Current_Reactive.Power_kVarh	Leading_Current_Reactive_Power_kVarh	CO2(tCO2)	Lagging_Current_Power_Factor	Leading_Current_Power_Factor	NSM	WeekStatus	Day_of_week	Load_Type
0	01/01/2018 00:15	3.17	2.95	0.0	0.0	73.21	100.0	900	Weekday	Monday	Light_Load
1	01/01/2018 00:30	4.00	4.46	0.0	0.0	66.77	100.0	1800	Weekday	Monday	Light_Load
2	01/01/2018 00:45	3.24	3.28	0.0	0.0	70.28	100.0	2700	Weekday	Monday	Light_Load
3	01/01/2018 01:00	3.31	3.56	0.0	0.0	68.09	100.0	3600	Weekday	Monday	Light_Load
4	01/01/2018 01:15	3.82	4.50	0.0	0.0	64.72	100.0	4500	Weekday	Monday	Light_Load



Data Profile: Analysed **35,040** rows of operational metrics from the steel industry dataset.

Data Cleaning: Implemented statistical profiling to identify and handle extreme consumption outliers (Usage_kWh).

Introduction to Regression Evaluation:

Regression: It is a way to predict an exact number—like guessing the factory's exact electricity usage in kilowatts (kWh).

How do we know if our guess is good? We use simple mathematical scores to check the errors.

The main goal: We want to get our error scores as close to zero as possible, meaning our guesses are almost perfect.

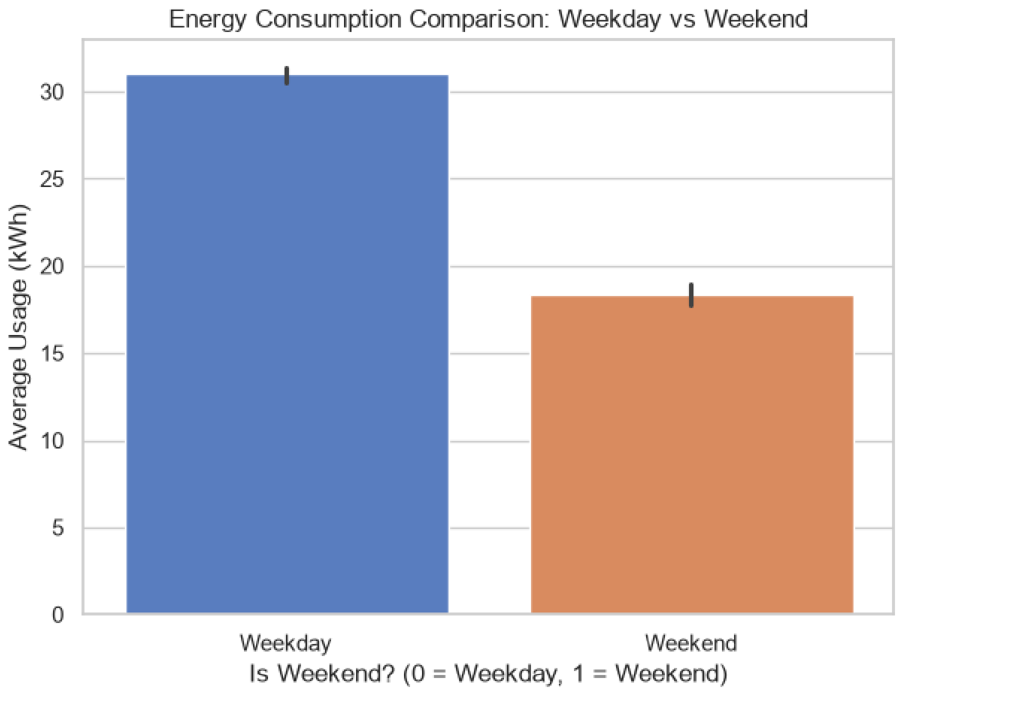
Our Tech Stack:

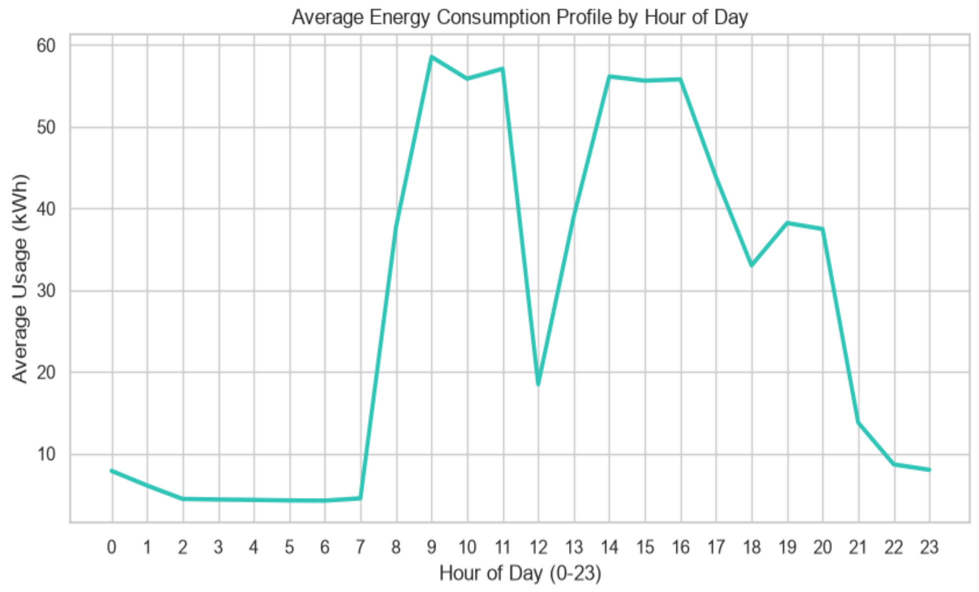
Language: Python (the core programming language used to write our code scripts).

Data Tools: Pandas and NumPy to load, clean, and organize our factory data tables.

Machine Learning Tool: Scikit-Learn (Sklearn) to quickly build and test our Linear Regression and Decision Tree models.

Temporal Energy Demand Patterns:





Temporal Cycles: Analysed average energy patterns across a 24-hour manufacturing cycle to map peak operational shifts.

Weekly Load Variance: Isolated a significant drop in power demands during weekends (averaging ~18 kWh) compared to heavy weekday production runs (averaging ~31 kWh).

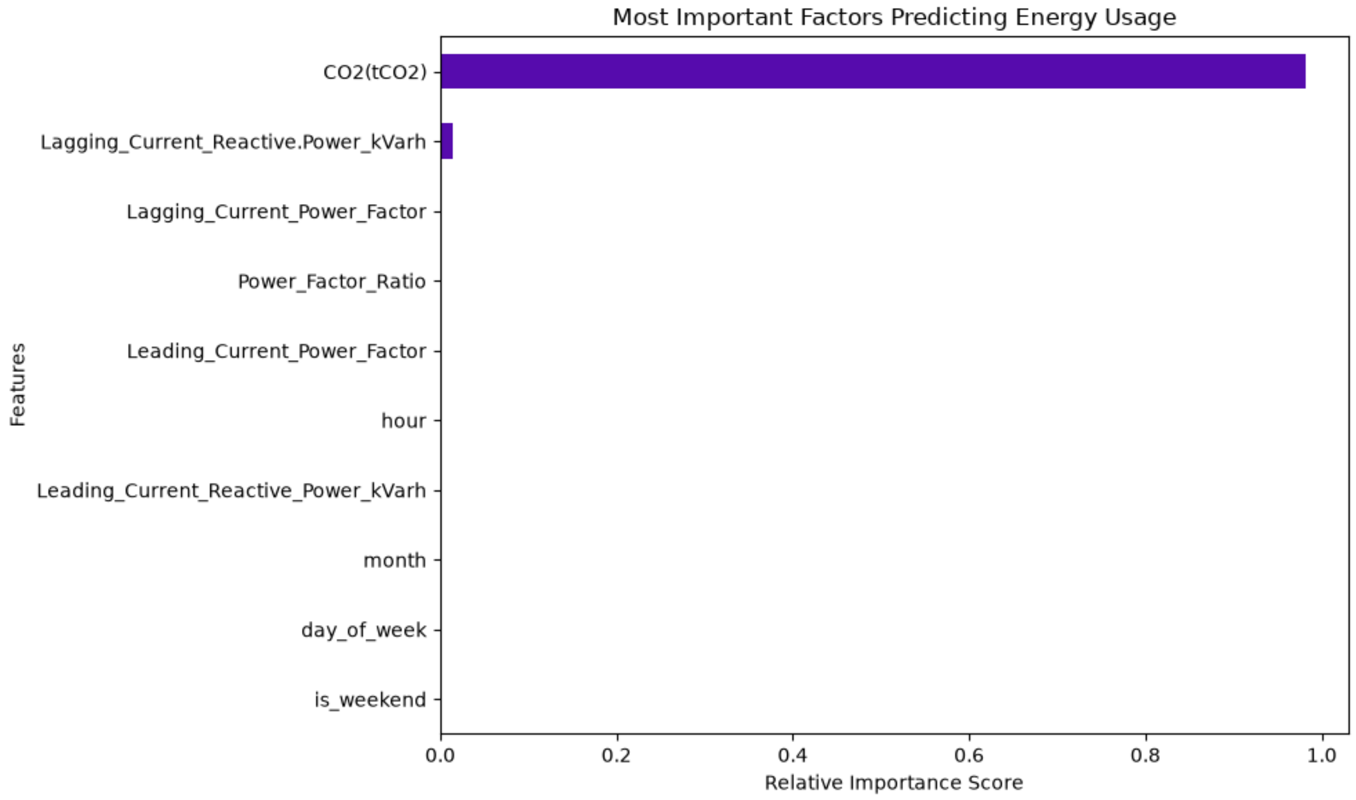
Regression Objective: Tasked models with learning these stark baseline shifts between operational days and hours to forecast grid demand.

Model Training & Initial Results: Feature Selection & Split:

Selecting Features: We chose the most important columns in our dataset (like power factors and carbon emissions) that directly affect how much electricity the factory uses.

Splitting the Data: We divided our dataset into a Training Set (to let the models learn the data patterns) and a Testing Set (to check how well they guess on new data).

Training the Models: We fed the training data into both the Linear Regression and Decision Tree systems so they could learn how to make predictions.



Feature Importance: Evaluated how features impact the model post-split.

Key Discovery: Carbon dioxide emissions (CO2(tCO2)) hold a relative importance score near 1.0, acting as the ultimate predictor for energy burn.

Model Performance Metrics:

Linear Regression Results:

MAE: ~3.49 kWh (On average, its guesses were off by about 3.5 units)

R^2 Score: ~0.67 (It understands about 67% of the data patterns)

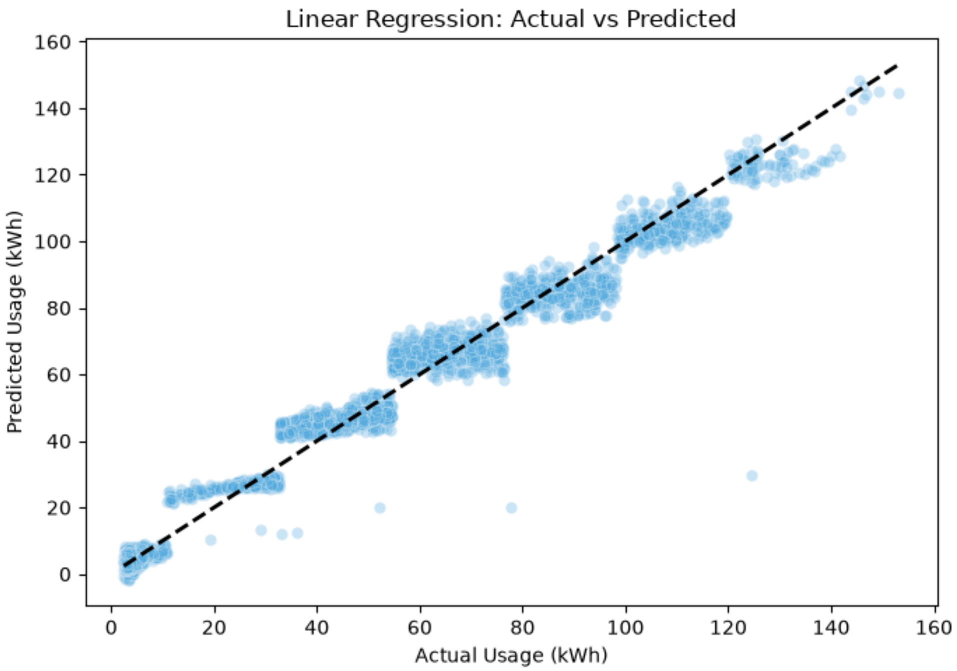
Decision Tree Results:

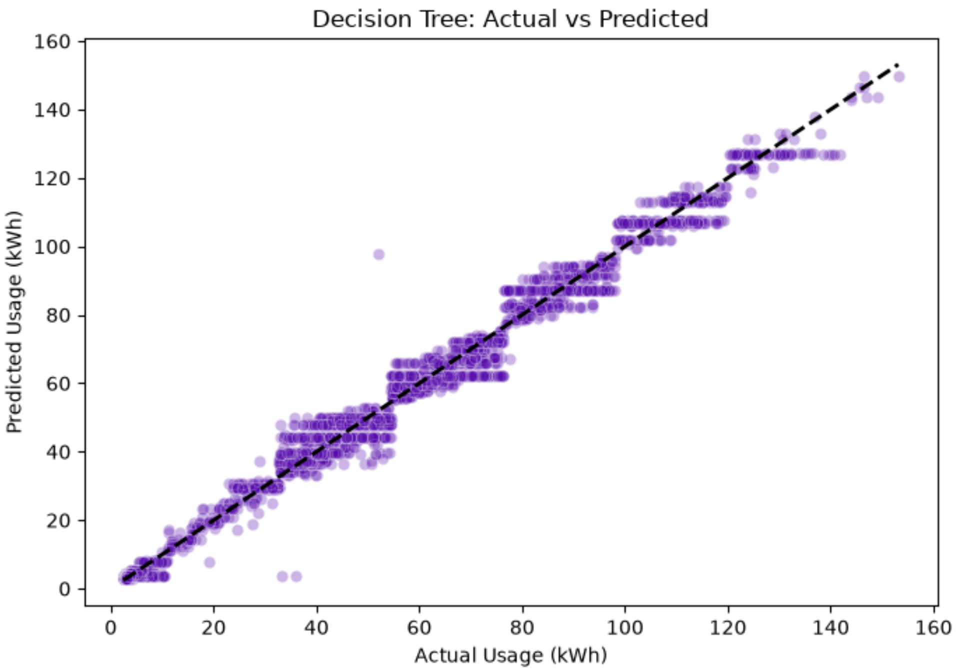
MAE: ~0.26 kWh (On average, its guesses were off by less than 1 unit)

R^2 Score: ~0.99 (It understands about 99% of the data patterns)

The Winner: The Decision Tree Regressor performed significantly better because it has a much lower error rate and a near-perfect prediction score.

Predicted vs Actual Usage (kwh):





Conclusion & Next Steps: Core Achievements:

Conclusion: We successfully set up our baseline modelling environment. The Decision Tree model gave us a fantastic starting benchmark with an R^2 score of 99%.

What I Learned: I learned how to split data into training and testing sets, how to use Scikit-Learn to build models, and how to read error scores like MAE.

Next Steps for Week 3: Moving forward, we will explore hyperparameter tuning to optimize our models further and look into advanced algorithms to see if we can improve our predictions even more.